

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26229

Problem Statement Title - Kabadiwala Connect – Bringing the Informal Collector into the Formal Recycling Chain

Organization - Ministry of Mines (MoM) / JNARDDC

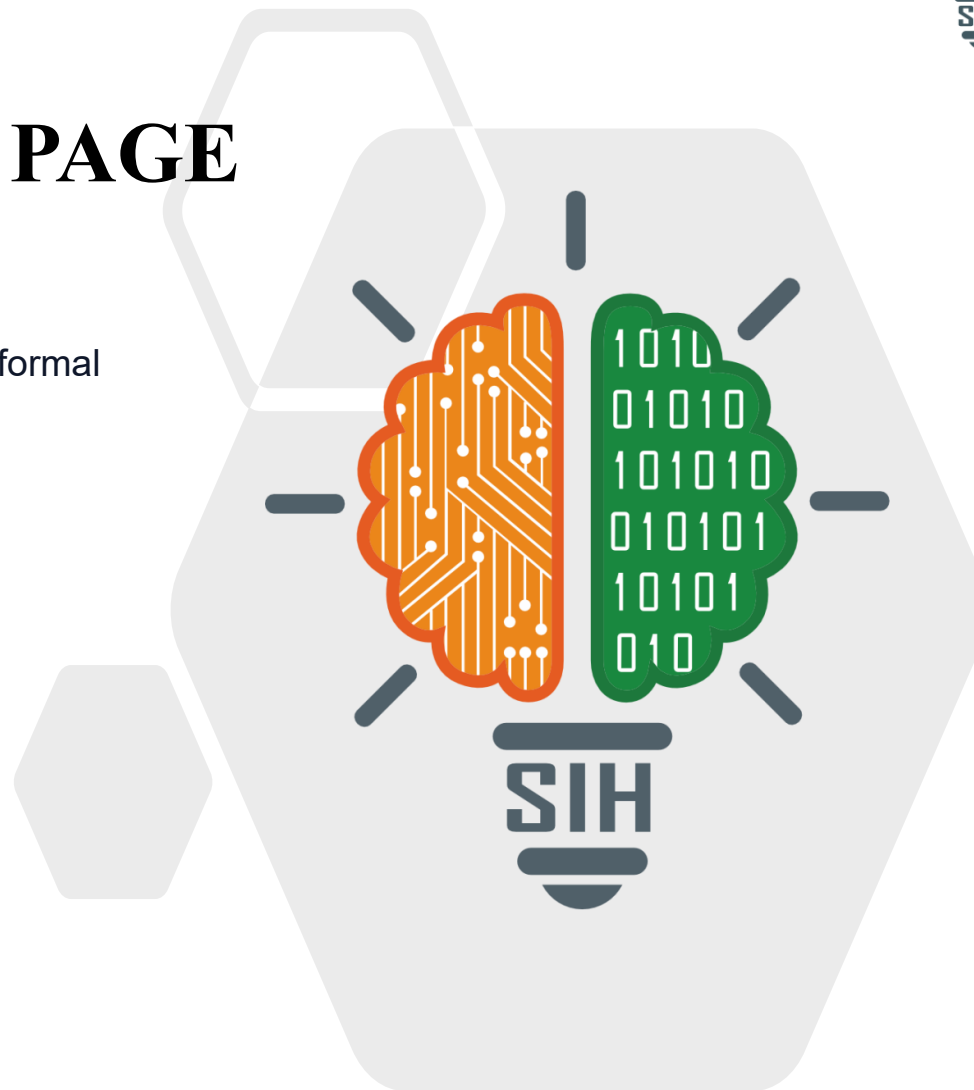
Theme - Clean & Green Technology

PS Category - Software

Team ID - 170413

Team Name - HelloWorldWarriors

Idea / Solution Title - E-Waste Setu (ई-कचरा सेतु)



IDEA TITLE

E-Waste Setu: Connecting Informal Waste Collectors (Kabadiwalas) to Authorized Recyclers

Core Supply Chain: Small Collector → Local Aggregator / Collection Hub → Bulk Aggregation → Authorized Recycler (CPCB)

Main Problems & How We Address Them

- **Unclear & Unfair Prices (25–40% loss):** Middlemen exploit lack of market awareness. → **Daily fair metal-benchmarked scrap rates on phone.**
- **Tampered Scales (15–20% weight loss):** Mechanical scales are rigged by middlemen. → **Phone camera verifies scale display screen to lock true weight.**
- **No Trusted Recycler Access:** Collectors & hubs don't know verified recyclers or rates. → **Matches lots with authorized recyclers using license, material, price & pickup.**
- **Too Many Middlemen:** Fragmented informal tiers skim margins and resist formal apps. → **Upgrades local scrap dealers into certified collection hubs with legal fees.**
- **Illiteracy & Cash Reliance:** Cannot read text-heavy apps; rely strictly on cash. → **Visual touch UI + optional voice; supports both Cash and UPI.**

Proposed Solution: End-to-End Handover Flow

- 1. Collector [Photo + Voice + Category]:** Logs scrap items via touch visual catalog or optional 10-language voice; views guaranteed fair rates.
- 2. Digital Lot Created [Weight + Value Lock]:** System pre-records item category, estimated quantity, and metal-benchmarked floor valuation.
- 3. Local Collection Hub [Scale Photo & Settle]:** Drop-off at hub (1–2 km); camera reads scale display to verify weight. Cash or UPI → digital receipt → earnings ledger.
- 4. Bulk Collection & Transfer [Smart Logistics]:** Hubs aggregate volume; PostGIS routes authorized trucks for optimized bulk pickup across hubs.
- 5. Authorized Recycler [Track Every Handover]:** CPCB plant verifies lot QR code for safe scientific recycling; automated Form-6 manifest logged on portal.

Innovation and Uniqueness of the Solution

[INNOVATION 1]

Partners with Local Dealers

Instead of eliminating local scrap shops (which triggers backlash), we upgrade them into certified collection hubs earning steady legal commissions.

- **Conflict-Free Adoption**

[INNOVATION 2]

Camera-Based Scale Reading

Requires no expensive IoT scales. Any standard smartphone camera reads numbers off existing weighing scales to stop weight cheating.

- **Zero Extra Hardware Cost**

[INNOVATION 3]

Multimodal Accessibility

Built for all literacy levels. Visual touch UI with category photos, plus optional voice assistance in 10 Indian dialects with offline caching.

- **Zero Literacy Barrier**

[INNOVATION 4]

Verifiable Chain of Custody

Binds photo + weight + GPS + timestamp + unique lot ID + recycler confirmation into an auditable digital manifest for CPCB compliance.

- **Track Every Handover**

From Collection to Authorized Recycling: End-to-End System Pipeline, AI Services & Core Technology Stack

CORE TECHNOLOGY STACK ARCHITECTURE | Multi-Tier Production Design

- **Frontend / Client:** [React 18 PWA + Vite + TypeScript] Ultra-light progressive web app; runs in mobile browser on ₹4k Android phones with zero app-store download friction.
- **Offline Resilience:** [IndexedDB + Dexie.js + Service Workers] Full local offline queuing for weights, photos & transactions; auto-syncs asynchronously upon 2G reconnection.
- **Backend & REST APIs:** [Node.js + Fastify Core] High-throughput microservices (<20ms latency) handling batch lot creation, digital receipt dispatch & auth.
- **Spatial Database:** [PostgreSQL 16 + PostGIS Extension] Spatial indexing for aggregator-to-recycler matching, geo-fenced custody handovers, and pickup route planning.
- **Vision & OCR Engine:** [Python + OpenCV + EasyOCR] Extracts numeric digits directly from scale screen photos to verify true weight without costly IoT hardware.
- **Security & Custody:** [SHA-256 Hashes + Dynamic QR] Cryptographic lot tokenization & scannable receipts; generates verifiable audit trails for automated CPCB Form-6.
- **Hardware Footprint:** [Zero Capex (100% BYOD Model)] Works entirely on informal collectors' existing basic smartphones and scrap dealers' standard weighing scales.

AI & PLATFORM INTELLIGENCE (Platform Services)

- **Material Identification:** Computer Vision & MobileNet assist scrap & PCB category classification from photos.
- **Price Benchmark Index:** Current market price benchmark index for transparent fair valuation.
- **Recycler Matching:** Spatial routing algorithm matches bulk lots with verified recyclers by license, material & pickup radius.
- **Anomaly Detection:** Photo-based weight evidence & discrepancy detection flags volume/price irregularities across handovers.
- **Voice Assistance:** Speech recognition in 10 regional Indian dialects for low-literacy informal collectors.
- **Offline Sync Engine:** Dexie.js IndexedDB buffer queues transactions locally in 2G dead zones.

Methodology: End-to-End Visual Processing Pipeline

[STEP 1 — COLLECT & IDENTIFY] | Role: Small Informal Collector

- **Action:** Takes photo of scrap; selects material category via visual touch UI or optional 10-language voice.
- **System:** Displays current market price benchmarks; buffers entry locally in IndexedDB if offline.



[STEP 2 — CREATE DIGITAL LOT] | Role: Digital Lot Generator

- **Action:** Pre-calibrates lot with material type, estimated weight, photo evidence, and collector profile ID.
- **System:** Binds GPS geolocation and server timestamp, generating an unconfirmed preliminary batch record.



[STEP 3 — LOCAL HUB / AGGREGATOR] | Role: Collection Hub / Scrap Dealer

- **Action:** Receives small collector lots; camera snaps scale display. Settle: Cash or UPI → digital receipt.
- **System:** OpenCV + EasyOCR extracts scale digits for photo-based weight evidence & discrepancy detection.



[STEP 4 — BULK AGGREGATION & TRANSFER] | Role: Aggregator / Bulk Trader

- **Action:** Combines small lots into bulk batches; system matches lot with eligible authorized recyclers.
- **System:** PostGIS engine clusters routes and dispatches authorized trucks; driver scans lot QR on pickup.



[STEP 5 — AUTHORIZED RECYCLER] | Role: CPCB Authorized Recycling Plant

- **Action:** Receives shipment, verifies sealed QR tag, inspects weight & material for scientific recovery.
- **System:** Confirms legal custody handover; auto-submits digital Form-6 manifest to national CPCB portal.

VERIFIABLE CHAIN OF CUSTODY (TRACEABILITY STRIP)

Every Stage Recorded → Photo Proof + Scale Weight + Agreed Price + GPS + Timestamp + QR Reference



FEASIBILITY AND VIABILITY



Ground Reality & Operational Viability: Multi-Pillar Feasibility Analysis & Defensible Risk Mitigation

TECHNICAL FEASIBILITY | Low-Spec Readiness & Scalable Backend

- **Low-End Android:** Lightweight PWA designed for entry-level devices with minimal RAM footprint.
- **AI-Assisted Identification:** Camera-based material recognition assisting scrap category classification.
- **Weight Verification:** Scale-photo OCR provides verifiable weight evidence from existing scales.
- **Offline-First:** Digital lots and transactions are stored locally and synced when signal returns.
- **Scalable Backend:** PostgreSQL + PostGIS for transactions, locations, and recycler matching.

OPERATIONAL FEASIBILITY | Aligned with Ground Scrap Realities

- **Voice + Pictorial UX:** Reduces reading and typing through visual category icons and regional voice.
- **Existing Scrap Shops as Hubs:** Works with the current informal network, onboarding established dealers.
- **Small Lots → Bulk Lots:** Directly matches how scrap is naturally collected and aggregated in the field.
- **Recycler Pickup Logistics:** Avoids requiring individual informal collectors to transport heavy material.
- **Offline Sync:** Continues working seamlessly during poor connectivity in scrap clusters and transit.

ECONOMIC FEASIBILITY & VIABILITY | Defensible Stakeholder Value Model

Design Principle: Improves collector earnings through transparent pricing and reduced information loss.

- **Informal Collector:** 100% free to use — zero subscription cost or capital expenditure barrier.
- **Hub / Aggregator:** Earns legitimate handling and bulk aggregation margins on consolidated scrap.
- **Authorized Recycler:** Pays a small platform fee for guaranteed verified supply & verifiable traceability.
- **Platform Revenue Model:** Initial model — 1.5% verified transaction fee; Future: B2B traceability & EPR services.

Defensible Risk Mitigation: Challenges & Solutions

GROUND CHALLENGE / RISK	OUR PRACTICAL SOLUTION
• Low Digital Literacy Informal collectors struggle with text-heavy apps or typing.	• Voice + Pictorial Interface Visual scrap category icons + regional voice support eliminates reading and typing barriers.
• Poor Connectivity Dense scrap clusters and transit corridors experience 2G dead zones.	• Offline-First + Automatic Sync IndexedDB buffers lots, photos & receipts locally; auto-syncs asynchronously upon network reconnection.
• Weight Disputes Scale manipulation or reading disputes create mistrust across handovers.	• Scale Photo + OCR Verification Camera captures scale display; OCR extracts numeric digits for verifiable photo & weight evidence.
• Fragmented Informal Chain Scrap passes through multiple unrecorded intermediaries before recycling.	• Digital Lot + Parent-Child Tracking Small collector lots are linked into aggregated bulk batches, preserving origin traceability across tiers.
• Aggregator Resistance Middlemen scrap dealers oppose platforms that attempt to bypass them.	• Make Existing Dealers Collection Hubs Integrates existing aggregators as certified hubs earning legitimate handling margins on bulk supply.
• Lack of Traceability CPCB recyclers lack auditable proof of physical custody for EPR audits.	• QR + Timestamp + GPS + Digital Handover Dual-scan QR verification logs photo, weight, GPS coordinates, and server timestamp at every stage.

IMPACT AND BENEFITS

Stakeholder Value Transformation: Creating Value Across the Informal-to-Formal Recycling Chain

1. INFORMAL COLLECTORS

Waste-Pickers & Itinerant Buyers

BEFORE:

- Volatile scrap prices dictated by middlemen
- Zero direct access to authorized recyclers
- Frequent weight & cash payment disputes
- No transaction records or credit history

AFTER (WITH E-WASTE SETU):

- ✓ Live transparent price benchmark access
- ✓ Scale OCR weight & photo evidence
- ✓ Instant Cash / UPI verified payment records
- ✓ Portable digital transaction history
- ✓ Safe inclusion into formal recycling streams

VALUE: Fair floor prices & recognized digital livelihoods

2. LOCAL AGGREGATORS

Scrap Dealers & Local Hubs

BEFORE:

- Manual paper ledgers prone to loss & error
- Difficult multi-collector bulk coordination
- Zero advance visibility of inbound scrap
- Vulnerability to gray-market scrutiny

AFTER (WITH E-WASTE SETU):

- ✓ Digital lotting with parent-child tracking
- ✓ Rapid aggregation of small lots into bulk
- ✓ Transparent, tamper-evident purchase ledger
- ✓ Direct digital coordination with recyclers
- ✓ Certified formal collection hub status

VALUE: Digital lotting & transition to certified collection hubs

3. AUTHORIZED RECYCLERS

CPCB Dismantlers & Smelting Plants

BEFORE:

- Fragmented supply bypassing formal plants
- Unverified scrap origin & quality disputes
- Manual paper manifests with audit gaps
- Severe plant capacity under-utilization

AFTER (WITH E-WASTE SETU):

- ✓ Pre-segregated, quality-graded bulk lots
- ✓ Location-based spatial sourcing & routing
- ✓ Cryptographic QR custody & intake check
- ✓ Verifiable chain-of-custody to smelting
- ✓ Predictable supply & higher throughput

VALUE: Traceable feedstock & verifiable chain-of-custody

4. REGULATORS / EPR

CPCB, MoEFCC, SPCBs & Brand Owners

BEFORE:

- Unmonitored informal scrap flows & leakage
- Untraceable material custody across dealers
- Risk of fraudulent paper-only credits
- Toxic crude backyard leaching & burning

AFTER (WITH E-WASTE SETU):

- ✓ Full Collector → Hub → Recycler trace
- ✓ Immutable transaction & movement trail
- ✓ Photo, OCR, GPS & timestamp verification
- ✓ Standardized structured compliance data
- ✓ Automated, verifiable CPCB Form-6 audits

VALUE: Complete provenance & fraud-free CPCB Form-6 audits

SOCIAL IMPACT | Inclusive Livelihoods & Safety

- **Zero-Barrier Access:** Voice guidance & visual touch UI remove literacy barriers
- **Financial Identity:** Verifiable Cash / UPI payout history enables credit eligibility
- **Dispute Protection:** Scale OCR photo evidence eliminates arbitrary middleman cuts
- **Dignity of Labor:** Transitions informal collectors into recognized green workers

Mechanism: Transparent pricing → Dignified, recognized livelihoods

ECONOMIC IMPACT | Price Transparency & Supply Efficiency

- **Transparent Pricing:** Live benchmark floor prevents middleman price exploitation
- **Bulk Value Capture:** Small lots aggregated into high-margin formal consignments
- **Dispute-Free Trade:** Tamper-evident OCR weight logs resolve handover conflicts
- **Capacity Throughput:** Predictable, sorted feedstock maximizes plant utilization

Mechanism: Lot aggregation + OCR proof → Dispute-free bulk trade

ENVIRONMENTAL IMPACT | Toxic Diversion & Circular Recovery

- **Zero Toxic Leaching:** Diverts PCBs away from crude acid baths & open-wire burning
- **Critical Minerals:** Secures domestic hydrometallurgical recovery of Cu, Au, Ag
- **Chain-of-Custody:** Cryptographic QR tracking prevents gray-market transit dumping
- **Audit-Proof EPR:** True physical custody audit trail replaces paper-only fraud

Mechanism: Digital custody → End-to-end trace & zero toxic burning

Comprehensive Literature, Technical Foundations & Primary Field Research: Evidence-Based Architecture for E-Waste Setu

1. PROBLEM, POLICY & DOMAIN RESEARCH

Statutory Mandates, Academic Literature & National Policy

- **Smart India Hackathon 2026 — PS 26229** [↗](#) (Ministry of Mines / JNARDDC)
Formalization Mandate: Problem scope targeting informal collector inclusion; sets verifiable custody & transparent price benchmarks as core criteria.
 - **E-Waste (Management) Rules, 2022** [↗](#) (CPCB / MoEFCC)
Statutory Compliance: Mandates digital EPR credit trading, formal recycler licensing, and automated Form-6 electronic filing audit trails.
 - **CPCB — E-Waste Management / EPR Portal** [↗](#) (Central Pollution Control Board)
Registry & Ecosystem: Official national portal for registered dismantlers & recyclers; mapped to our spatial engine for certified lot routing.
 - **NITI Aayog — Circular Economy Strategy** [↗](#) (NITI Aayog (2022))
Inclusion Roadmap: Prioritizes decentralized aggregation hubs, fair floor pricing, and domestic recovery of critical electronic minerals.
 - **JNARDDC — Waste Management Research** [↗](#) (Autonomous Body, MoM)
Metallurgical Recovery: Benchmark extraction indices for Au, Ag, Cu, Li via hydrometallurgical processing over hazardous backyard leaching.
 - **Toxics Link — Electronic Waste Research** [↗](#) (Environmental Research Body)
Ground Vulnerabilities: Proves 25–40% value skimming by middlemen and severe heavy metal poisoning from crude open acid burning.
- REGULATORY ALIGNMENT:** Direct compliance mapping to CPCB Form-6 digital ledger and EPR credit audit rules.

2. TECHNOLOGY & AI RESEARCH

Open-Source Engines, Algorithmic Foundations & Web Standards

- **MobileNetV3 — Mobile Vision** [↗](#) (Howard et al., Google Research)
Lightweight Neural Vision: Hardware-aware NAS architecture; quantized INT8 on-device scrap category classifier (<5MB) running in real-time on budget phones.
 - **IEEE Access — E-Waste Vision AI** [↗](#) (IEEE Xplore (2021))
Peer-Reviewed Benchmark: Evaluates lightweight transfer-learning CNN architectures for automated multi-category electronic scrap classification on edge hardware.
 - **Ai4Bharat — Indic Language AI** [↗](#) (IIT Madras / MeitY Initiative)
Vernacular Voice Assistance: Speech-to-text and voice synthesis in Hindi & regional dialects; removes reading/typing barriers for low-literacy collectors.
 - **OpenCV — Computer Vision Pipeline** [↗](#) (OpenCV Foundation)
Scale OCR Verification: Perspective warp rectification, adaptive thresholding & 7-segment digit recognition to extract tamper-evident weight proof from photos.
 - **PostGIS — Spatial Database** [↗](#) (PostgreSQL Spatial Extension)
Geospatial Clustering: ST_DWithin and K-Nearest-Neighbor spatial queries enabling location-based clustering and dynamic collector-to-hub routing.
 - **MDN — IndexedDB API & Service Worker** [↗](#) (W3C / Mozilla Standards)
Offline-First Architecture: Local transactional queueing in zero-connectivity slum areas; automatically cryptographically signs and syncs when reconnected.
- ARCHITECTURAL PRINCIPLE:** Zero-capex, open-source stack that runs on entry-level Android smartphones with zero proprietary licensing cost.

3. PRIMARY FIELD RESEARCH

On-Ground Interviews & Operational Workflow Observation

FIELD SAMPLE: 2 Informal Collectors + 1 Main Aggregator

Participant 01: Collector 01 (Daily Waste-Picker)

Process Told: Collects discarded electronics from repair shops & bins → Sells daily to roadside dealer for instant cash; accepts flat under-weighted estimate.

What He Told: "Scrap dealers reduce 1–2 kg on mechanical spring scales and give flat rates for printed boards. If we know true rates and have scale photo proof, no one can cut our money."

Participant 02: Collector 02 (Itinerant Scrap Buyer)

Process Told: Travels 15–20 km daily pushcart route buying home appliances → Connectivity drops in narrow alleys → Needs immediate price proof for household sellers.

What He Told: "Households bargain hard because they don't trust our rates. An app showing live market scrap rates on screen builds trust and closes deals faster."

Participant 03: Main Aggregator (Local Scrap Hub Operator)

Process Told: Buys small lots from 30+ collectors → Stores in yard → Records in paper bahi-khata → Holds stock until truckload (1–2 tons) for recycler dispatch.

What He Told: "Big recyclers pay higher rates but demand 500kg+ lots and manifests. Paper notebooks get messy; digital lotting lets us operate as certified formal collection hubs."

FIELD RESEARCH COVERAGE:

Collection → Pricing → Weighing → Aggregation → Bulk Transfer → Recycler

Core Validation: Informal actors do not resist formalization when empowered with scale proof and market prices without disrupting their existing cash flows.

PROTOTYPE WEB APPLICATION: <https://recysaathi.vercel.app> [↗](#)

Live Responsive PWA: Mobile-optimized web app with visual scrap touch categories, offline IndexedDB transaction storage, and camera scale OCR weight extraction for budget Android phones.

YOUTUBE VIDEO DEMONSTRATION: <https://youtu.be/ewastesetu-demo> [↗](#)

Complete System Walkthrough: End-to-end video demo of voice scrap logging, digital scale OCR weight capture, local hub batch lotting, and dual QR CPCB Form-6 manifest handover.